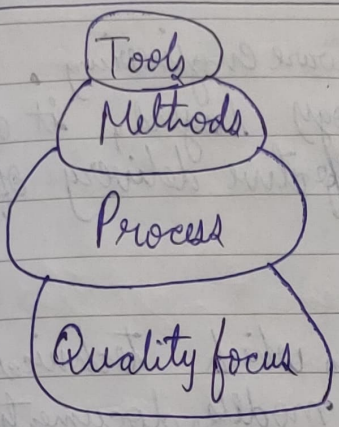
* **Software** = Computer + Data + Documents:
  Program Structure Soft & Hard

- Computer program that when executed provides desired features, functions & performance.
- Data structure that enable programs to easily manipulate information
- Descriptive information in both hard & soft copy that describes the operation & use of programs.

* **Characteristics**

- Software is developed or engineered, not manufactured
- Unlike hardware, no physical manufacturing process exists
- Cost is concentrated in development & maintainance, not production.
- Software does not wear out, unlike hardware, hardware follows a bathtub curve. Software does not physically degrade with time
- Failure occur mainly due to:
  - Requirement changes
  - Environment changes
  - Poor maintenance
- Software is intangible, cannot be touched or physically measured.

* Software Engineering Layered Approach



1) Tools

- Software Engineering tools allows automation of activities which helps to perform systematic activities. A system for the support of software development, called computer Aided Software Engineering (CASE)
- Software Engineering tools provide automated or semiautomated support for the process & the methods
- CASE – Computer Aided Software Engineering is Scientific ~~approach~~ application of a set of tools & methods to a software system which is meant to result in high quality, defect free & maintainable software products.
- CASE tools automate many of activities involved in various life cycle phases.

2) Methods

- It provides technical how-tos for building software, it encompasses many tasks including communication, requirement analysis, design modelling, program construction

3) Process

- It is foundation of Software engineering. It is glue that holds the technology layers, it defines a framework activities for effective delivery of software engineering technology.

- It establish the context in which technical methods are applied, work products (models, documents, data, reports, forms, etc) are produced, milestones are established, quality is ensured & change is properly managed.

4) Quality

- Main principle of software engineering is Quality focus.
- Engineering approach must have a focus on quality.
- Total Quality Management (TQM), Six Sigma, ISO 9001, ISO 9000-3, Capability Maturity Model (CMM), CMMI & similar approaches encourages a continuous process

॥ ॐ नमो भगवते वासुदेवाय नमः ॥

Unit-1 Background ESE Paper — 3/6/2024

Q1) (a) Program vs Software product — 3M

Program	Software product
• Program is a set of instructions written in a programming language to perform a specific task.	Software product is a complete software system that includes programs, documentation, data structures and other supporting components delivered to users.
• Limited scope, usually performs a single or specific task.	Broad scope, designed to solve real world problems for many users.
• Mainly consists of source code only.	Includes programs, documentations, manuals, data structures, installation guides and user interfaces.
• Usually developed by an individual or small team.	Developed using software engineering processes by organized teams.

4M

Q1)(b)

Prototype model	RAD Model
• A software development model where a working prototype of the system is built first to understand requirements & user feedback.	A model that focuses on rapid development of software using reusable components & parallel development.
• To clarify & refine user requirements through early prototypes	To develop software quickly & deliver it in a short time.
• An initial prototype is created then user feedback is collected then prototype is refined then final system is developed.	Development is divided into multiple modules that are developed simultaneously using tools & reusable components.
• Used when requirements are unclear or not well defined.	Used when requirements are well understood & stable.

Q1)(c) What is Software Engineering? Explain Software engineering a layered technology. —— 8 M

Ans 1)(c) Software Engineering is the application of engineered principles, methods & tools to develop high quality software in a systematic, disciplined & efficient manner.

It focuses on designing, developing, testing & maintaining software so that it is reliable, efficient & delivered within time & budget.

Software engineering aims to build software system that are:

- Delivered on time
- Developed within budget
- Have acceptable performance
- Provide correct & reliable operation.

* Software Engineering as a layered technology

Software Engineering is organized as a layered technology, where each layer supports the one above it. The layered architecture ensures quality, structure & systematic development of software. The layers are:

1. Quality focus
2. Process Layer
3. Methods Layer
4. Tools Layer.

1) Quality focus

- Base of software engineering
- It emphasizes continuous improvement of software quality
- Quality management practices such as TQM, Six Sigma, ISO standards & CMM support this layer.
- Without quality focus, software development becomes unreliable.

It ensures that the final software product meets required quality standards.

2) Process Layer

- The process layer provides the framework for software development.
- It defines the sequence of activities required to build software.
- Process activities include planning, design, coding, testing, requirement analysis, deployment & maintenance.

Purpose is to organize development work in a structured & manageable workflow.

3) Methods Layer

- Methods provide the technical techniques & procedures for building software.
- They include activities such as requirement analysis, system design, ~~etc~~ algorithmic design, coding, testing.

Purpose: To define how the software is actually built using engineering techniques.

4) Tools Layer

- Tools provide automated or semi-automated support for software development activities.
- These are known as CASE (Computer Aided Software Engineering) Tools.
- Examples are code editors, debugging tools, testing tools, bug tracking tools, version control systems (Git).
- Purpose is to increase productivity, accuracy & efficiency in software development.

→ Software Engineering is a systematic approach to software development that ensures reliable & high quality software.

The layered technology structure consists of

1) Quality Focus — foundation ensuring quality.
2) Process — framework of development activities
3) Methods — technical techniques to build software
4) Tools — automated support for development

Each layer supports the other, creating an organized & effective approach to building software systems.

Q2)(a) Purpose of timeline chart — 3M

Ans2)(a) ① Project Planning: Helps plan all project activities & arrange them in chronological order.

② Scheduling Tasks: Shows the start time & end time of each activity in the project.

③ Tracking Progress: Allows managers to monitor the progress of tasks & check whether the project is on schedule.

Q2)(a) Types of metrics — 3M

Ans2)(a) ① Process metrics: It measures the effectiveness & efficiency of the software development process. It improves the development process by identifying weakness in the process and supports long term process improvement.

② Project metrics: It measures the status & performance of a specific software project. Purpose is to track project progress, identify risks or delays, control cost & schedule. It helps managers monitor & control ongoing projects.

③ Product metrics: It measures the quality & characteristics of the software product itself. It evaluates software quality, measure complexity & reliability & ensures the product meets requirements.

Q2)(b) Agile Development — 4M

Ans2)(b) Agile Development is a software development methodology that focuses on iterative development, customer collaboration & rapid delivery of working software. It emphasizes flexibility so that changes in requirements can be handled easily during development.

In Agile, the software is developed in small increments called iterations or sprints & each iteration delivers a working part of the system.

Key characteristics :-

- Iterative Development : Project is divided into small cycles (iterations) & each cycle produces a working version of the software.
- Customer collaboration : Continuous interaction with customers ensures that the product meets user requirements.

Advantages :

- Faster delivery of software
- Better customer satisfaction
- Early detection of problems.

Q2)(b) List 5 functional requirements of Library Management System — 4M

Ans2)(b) Functional requirements define what the system should do & the services it must provide to users.

1) Book Management: The system should allow the librarian to add, update & delete & view the book records in the library.

2) Member Management: The system should register new members & maintain their details such as name, ID, contact information & membership status.

3) Book Issue fn: System should allow the librarian to issue books to registered members & record the issue date & due date.

4) Book Return fn: The system should enable the librarian to record the return of issued books & update the book availability status.

5) Fine calculation: The system should calculate & record fines automatically if a book is returned after the due date.

These functional requirements ensure that the library management system can manage books, members, transactions & fines efficiently.

Q2)(C) Activities in project planning? Error tracking? — 8M

Ans2)(C) Software project planning involves defining how the project will be executed, monitored & controlled.

1) Scope Definition: Clearly define project objectives, boundaries & deliverables.
Identify what is included & excluded in the project.

2) Estimation: Estimate cost, time & effort & resources required Based on past data & metrics.

3) Resource Planning: Identify & allocate human resources, tools & infrastructures.

4) Project Scheduling: Prepare schedules using Gantt charts & timeline charts. Define task sequences & deadlines.

5) Risk Management: Identify possible risks
Analyze & plan risk mitigation strategies

6) Task Allocation: Assign responsibilities to team members
Define roles clearly.

7) Monitoring & Tracking Plan: Defines how progress will be tracked. Use metrics to compare planned vs actual performance.

- Project planning activities ensures structured & efficient execution of software projects.

→ Error tracking is also known as bug tracking is the process of identifying, recording, managing & resolving errors (defects) in software during development & testing.

- ~~Error Id,~~ Each error is recorded with details like error id, description, Severity, status (open, in progress, fixed, closed).
- It helps developers track all defects systematically, assign bugs to team members, monitor progress of bug fixing, ensure no defect is missed.
- Common tools are JIRA, Bugzilla, Git Hub Issues.

→ Error tracking ensures Software quality by systematically managing & resolving defects.

Q) Characteristics of SRS (Software Requirement Specification) - 3M

Ans) 1) Correct: Every requirement should represent a real need of the user.

2) Unambiguous (clear): Each requirement should have only one interpretation.

3) Complete: All necessary requirements & details should be fully specified.

4) Consistent: Requirements should not conflict with each other.

5) Verifiable (Testable): Each requirement should be testable to check correctness.

6) Traceable to its source & related components.

Q) Software metrics — 3M

Ans) Software metrics are quantitative measures used to evaluate, control & improve different aspects of software development such as the process, product & project.

- Metrics provide numerical data about software.
- They help in planning, estimation, monitoring & quality control.
- Used to improve productivity & software quality over time.
- Essential for making data driven decisions ensuring efficient high quality software development.

Q) WSHH Principle — 4M

Ans) 1) Why is the system being developed?

- Identifies the business purpose & justification of project.

2) What will be done?

- Defines the tasks, scope & deliverables of the project.

3) When will it be done?

- Specifies project schedule & deadlines.

4) Who is responsible?

- Assigns roles & responsibilities to team members.

5) Where are they located?

- Identifies the organizational structure & team distribution.

6) How will the job be done?

- Describes the technical approach & development strategy.

7) How much is required?

- Estimates resources, cost & effort needed for the project.

Q) 4 P's of • Effective People Management — 3M

Ans) 1) People : Managers, developers, customers & users. team.
Focus on building a skilled, motivated & coordinated

2) Product : Defines what is to be developed. Includes project scope, objectives & requirements.

3) Process : Refers to framework or methodology used to develop the software. Includes activities like planning, design, coding, testing & deployment.

4) Project : Involves planning, scheduling, tracking & controlling the work. Ensures the project is completed on time, within budget & with required quality.